National University of Computer & Emerging Sciences, Karachi
Spring-2023 CS-Department
Final

June 2, 2023, Slot: 8:30 AM – 11:30 AM

Course Code: CS AI2002	Course Name: Artificial Intelligence
Instructor Name: Dr Fahad / Saeeda Kanwal / Dr.Waheed Ahmed/ Dr Muhammad Farrukh/ Dr.Farrukh	
Student Roll No:	Section No:

- Return the question paper.
- Read each question completely before answering it. There are **5 questions and 3 pages.**
- In case of any ambiguity, you may make assumption. But your assumption should not contradict with any statement in the question paper.
- All the answers must be solved according to the sequence given in the question paper.
- Be specific, to the point while coding, logic should be properly commented, and illustrate with diagram where necessary.

Time: **180 minutes.** **Max Marks**: **50 points**

Question 1 **CLO1 [10 Points]:**

Select the appropriate learning schemes of the mentioned applications in a table A below. Also justify your selection by giving two statements.

Table A Different Applications

Applications	Supervised Learning	Unsupervised Learning	Reinforcement Learning
Crowd monitoring in Hajj			
Gold (Jewellery) market prediction			
5G mobile communication			
Intelligent transport system			
Library			

Question 2 **CLO2 [10 Points]:**

A farm field is being covered by the unmanned aerial vehicle (UAV) to monitor the performance of several sensors deployed in the field. Assume that there are 5 temperature sensors being implemented in the southeast region of the field to determine temperature. The temperature values of the sensors are shown in the table below from 10:00 a.m. to 2:00 p.m. Assume that a linear regression machine learning model is implemented inside the UAV to predict the temperature correctly. Build a Linear Regression model based on the value given in table. B. Also calculate fitness of the developed model. Show all necessary steps.

Table. B Sensors and temperature Values.

Location	**Temperature Sensor ID**	**Temperature (in °C)**
Southeast	Sensor_Temp4HG	46.7
Southeast	Sensor_Temp5HG	47.8
Southeast	Sensor_Temp6HG	48.1
Southeast	Sensor_Temp7HG	48.2
Southeast	Sensor_Temp8HG	49.1

Question 3 **CLO3 [5+3+2 Points]:**

A. Given the following dataset: **(2, 10), (2, 5), (8, 4), (5, 8), (7, 5), (6, 4), (1, 2), (4, 9)**. Perform a k-Means clustering on this dataset. Initialize the centroids at (3, 7) and (9, 3). Use **Manhattan distance** as the distance metric. Solve for five iterations or until any other stopping condition is reached. List which points end up in which clusters.

B. How can you determine the optimal number of clusters (k) for a dataset in k-Means clustering? Explain the method briefly Using the elbow method. Plot the error for different values of k. Look for points where the error drops sharply or stops dropping sharply.

C. Can k-Means handle categorical or qualitative data? If YES, how? If NO, why?

Question 4 **CLO3 [3+7 Points]:**

A. Draw the Bayesian Network that corresponds to this conditional probability: P(D/A,B) P(E/D) P(B/A) P(C/A) P(C)

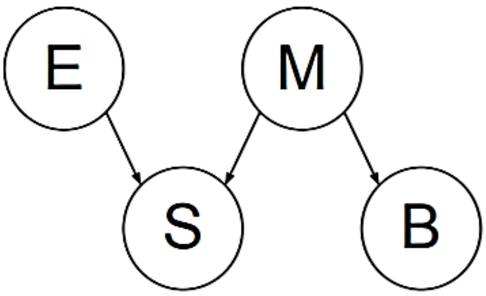
B. Compute the following for the figure shown below:

1: Compute joint Distribution P (E, ¬M, S, ¬B)?

2: Calculate P(+M| + S, +B, +E) using the network given above

$P(E)$	
$+e$	0.4
$-e$	0.6

$P(S\|E,M)$			
$+e$	$+m$	$+s$	1.0
$+e$	$+m$	$-s$	0.0
$+e$	$-m$	$+s$	0.8
$+e$	$-m$	$-s$	0.2
$-e$	$+m$	$+s$	0.3
$-e$	$+m$	$-s$	0.7
$-e$	$-m$	$+s$	0.1
$-e$	$-m$	$-s$	0.9



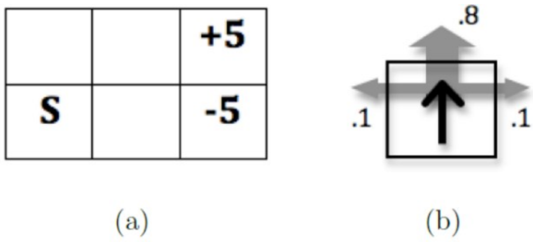
$P(M)$	
$+m$	0.1
$-m$	0.9

$P(B\|M)$		
$+m$	$+b$	1.0
$+m$	$-b$	0.0
$-m$	$+b$	0.1
$-m$	$-b$	0.9

Question 5 **CLO3 [2.5 + 2.5 + 5 Points]:**

$$V(s) = \max_a \left(R(s,a) + \gamma \sum_{s'} P(s,a,s')V(s') \right)$$

This grid world MDP operates like to the one we saw in class. The states are grid squares, identified by their row and column number (row first). The agent always starts in state (1, 1), marked with the letter S. There are two terminal goal states, (2, 3) with reward +5 and (1, 3) with reward -5. Rewards are 0 in non-terminal states. (The reward for a state is received as the agent moves into the state.) The transition function is such that the intended agent movement (Up, Down, Left, or Right) happens with probability .8. With probability .1 each, the agent ends up in one of the state's perpendicular to the intended direction. If a collision with a wall happens, the agent stays in the same state.

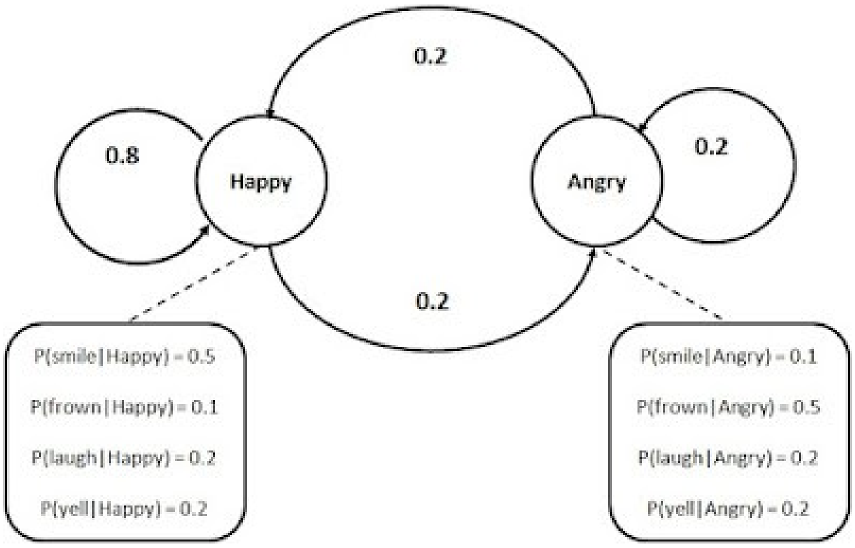


(a) Gridworld MDP. (b) Transition function.

1) Draw the optimal policy for this grid?
2) Transition probabilities and reward is already given in question. Apply the first two rounds of value iteration updates for each state, with a discount of 0.9. (Assume V0 is 0 everywhere and compute Vi for times i = 1; 2).
3) Suppose the agent does not know the transition probabilities. What does it need to be able do (or have available) to learn the optimal policy?

Question 6 **CLO3 [2 +2 + 2 + 4 Points]**

Mr. X is happy someday and angry on other days. We can only observe when he smiles, frowns, laughs, or yells but not his actual emotional state. Let us start on day 1 in the happy state. There can be only one state transition per day. It can be either to happy state or angry state. The HMM is shown below;



Assume that q_t is the state on day **t** and o_t is the observation on day t. Answer the following questions;

(a) What is P (q_2 = Happy)?

(b) What is P (o_2 = frown)?

(c) What is P (q_2 = Happy | o_2 = frown)?

(d) What is P (o_1 = frown o_2 = frown o_3 = frown o_4 = frown o_5 = frown, q_1 = Happy q_2 = Angry q_3 = Angry q_4 = Angry q_5 = Angry) if π = [0.7, 0.3]?

Good Luck !